ACL asosida paket filtrlash

ACL-Based Packet Filtering

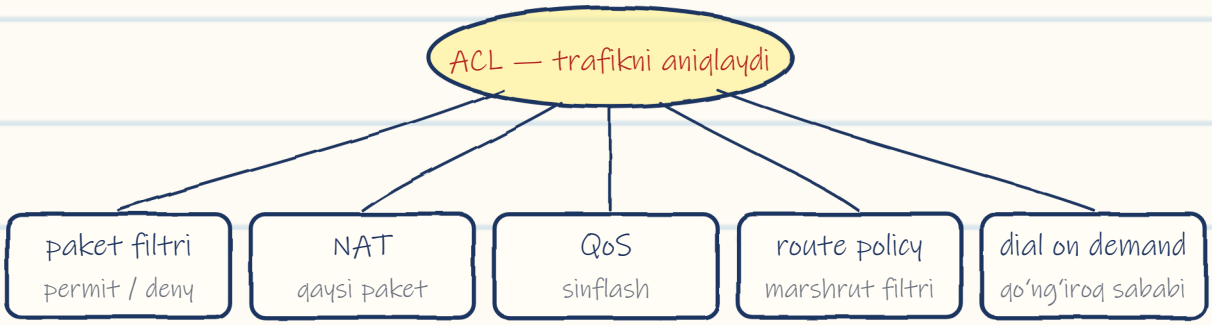
Asosiy atamalar

- ACL — paketni tanib olish qoidalari ketma-ketligi
- Qoida (rule) — bitta shart + amal: permit yoki deny
- Basic ACL — 2000–2999 — faqat manba IP bo'yicha tanlaydi
- Advanced ACL — 3000–3999 — manba/maqsad, protokol, port
- Wildcard — teskari maska: 0 — tekshir, 1 — tekshirma

1. ACL nima

Paketni tanib olish uchun qoidalar ketma-ketligi (Access Control List)

Har qoida = bitta amal (permit / deny) + moslash belgilari.



belgilar: manba/maqsad IP · protokol · port · MAC · VLAN

1-rasm. Bitta ACL — beshta qo'llanish

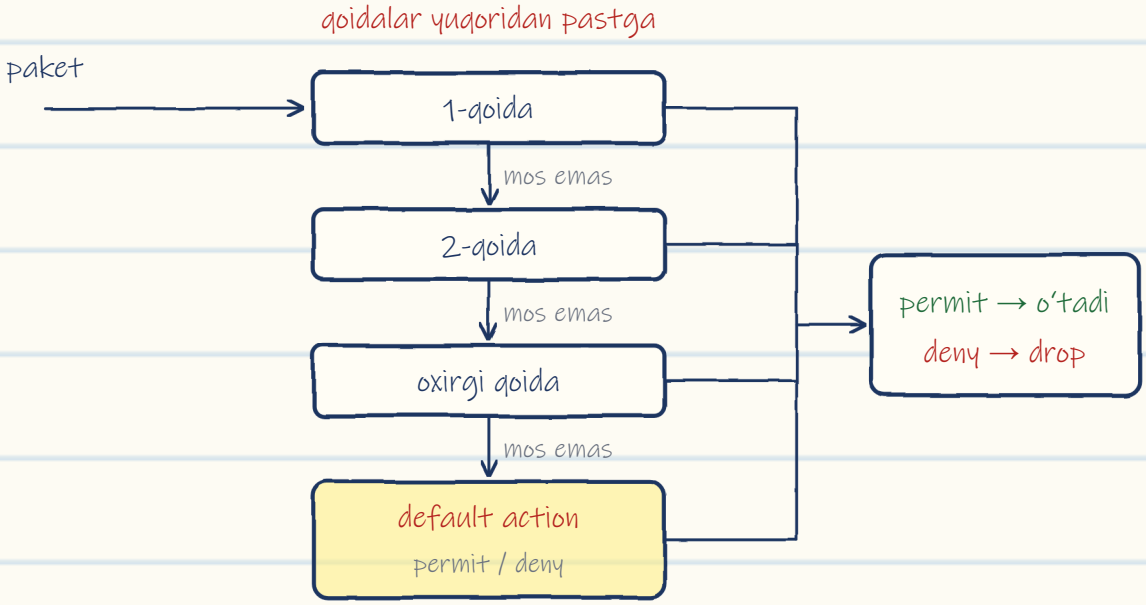
2. Interfeysda filtrlash



2-rasm. Har yo'nalishga alohida ACL: inbound va outbound

Yo'nalish: inbound — interfeysga kirgan paket, outbound — interfeysdan chiqayotgan paket.

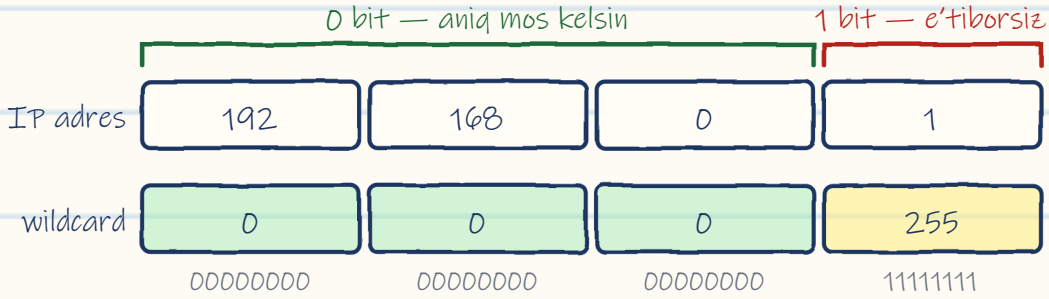
3. Qoidalarni tekshirish tartibi



3-rasm. Birinchi mos qoida ishlaydi — qolganlari tekshirilmaydi

- match-order config — rule ID o'sish tartibida (default)
- match-order auto — depth-first: IP diapazoni torroq qoida oldin
- Mos kelmadi → default action: firewall default permit | deny

4. Wildcard mask



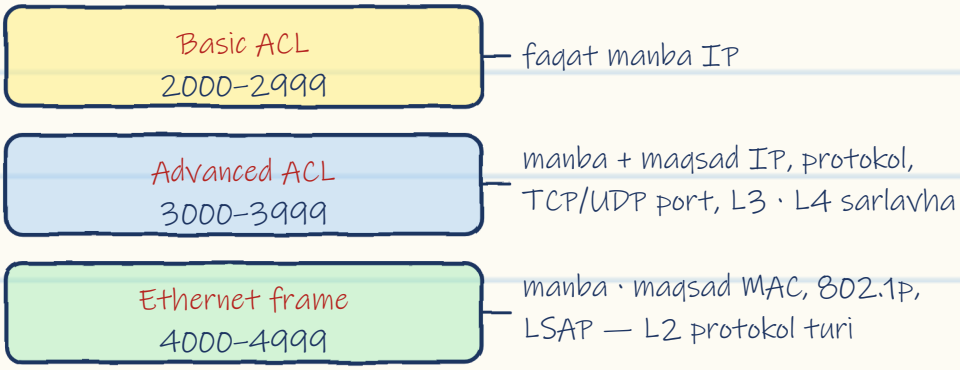
= 192.168.0.0/24 — faqat 1-24 bit tekshiriladi

1 bitlar ketma-ket bo'lishi shart emas

IP + wildcard = adres diapazoni:

IP	wildcard	diapazon
192.168.0.1	0.0.0.255	192.168.0.0/24
192.168.0.1	0.0.3.255	192.168.0.0/22
192.168.0.1	0.255.255.255	192.0.0.0/8
192.168.0.1	0.0.0.0	faqat 192.168.0.1
192.168.0.1	255.255.255.255	0.0.0.0/0 — hammasi
192.168.0.1	0.0.2.255	192.168.0.0/24 + 192.168.2.0/24

5. ACL turlari

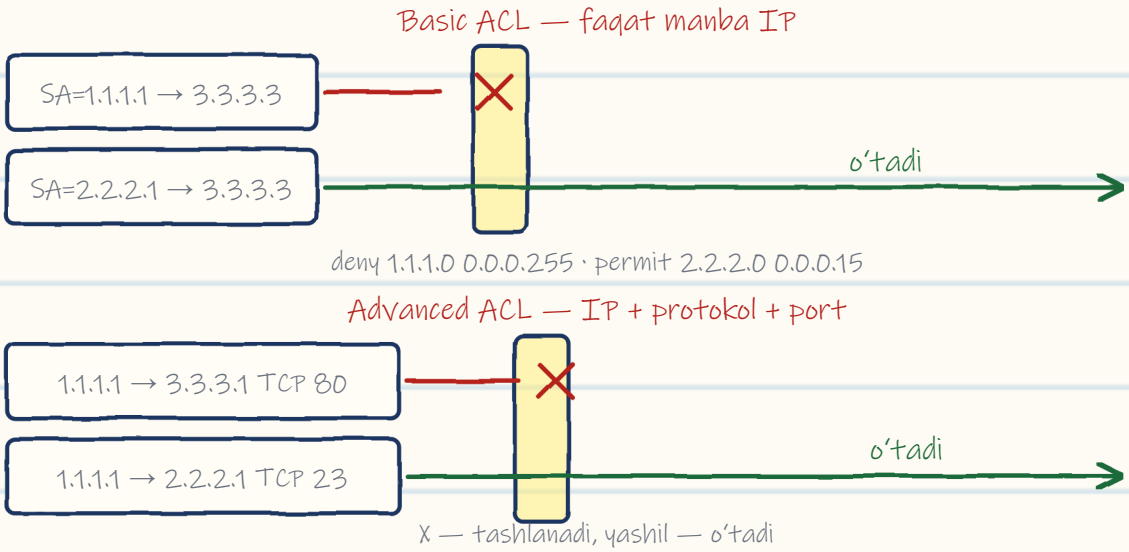


4-rasm. Raqam turni belgilaydi, tur esa moslash belgilarini

Sozlash:

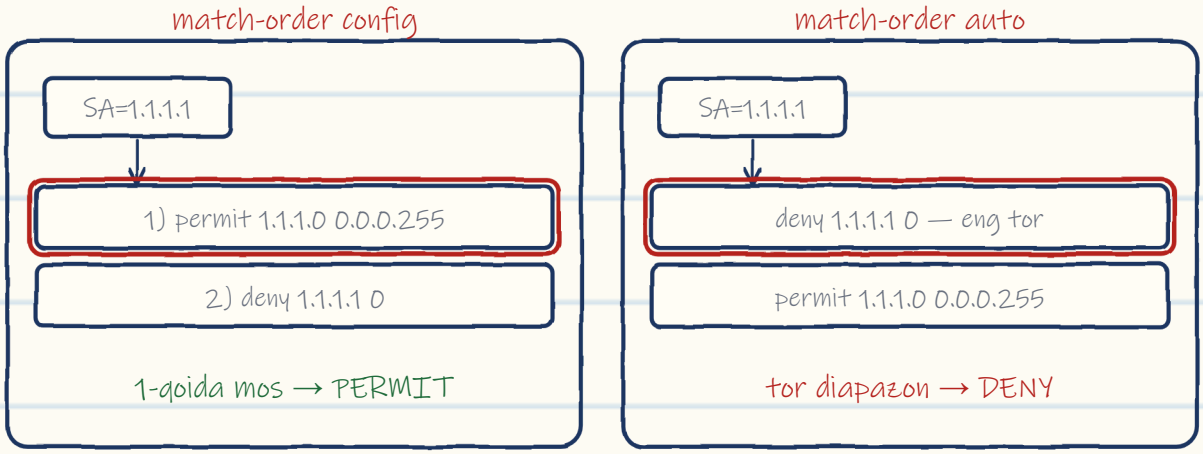
- acl number 2000 → rule permit source 10.0.0.0 0.0.0.255
- acl number 3000 → rule deny tcp source 1.1.1.0 0.0.0.255 destination 3.3.3.1 0 destination-port eq 80
- interfeysda: packet-filter 2000 inbound
- ko'rish: display acl 2001 · display packet-filter statistics

6. Basic va advanced — misol



5-rasm. Bir xil manba, lekin advanced ACL port va maqsadni ham ko'radi

7. config va auto — natija boshqa

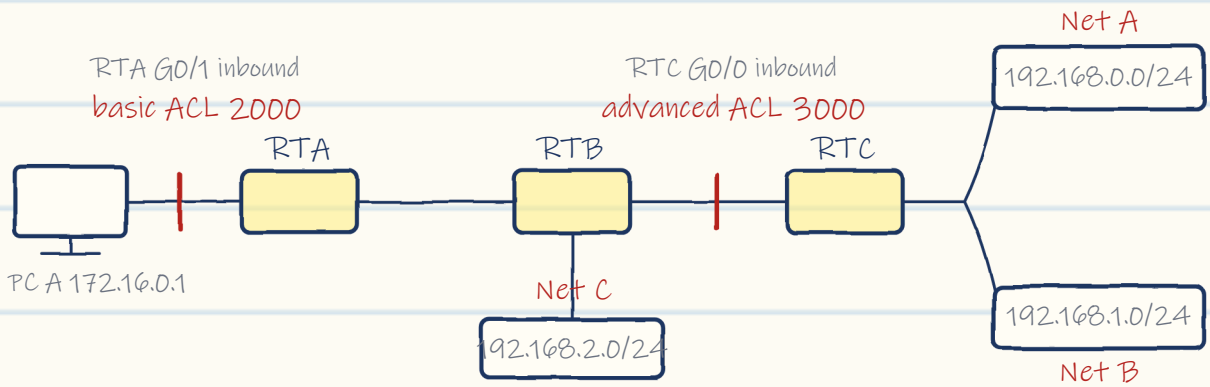


bir xil ACL, boshqa tartib — boshqa natija

6-rasm. Qizil ramka — birinchi tekshirilgan qoida

Sozlash tartibi: default action → ACL turini tanlash → qoidalarni yozish (belgi + permit/deny) → interfeysga qo'yish va yo'nalishni ko'rsatish.

8. ACL ni qaysi interfeysga qo'yish



PC A → Net A va Net B yopiq, qolganlari ochiq

7-rasm. Ikki yechim: RTA da basic yoki RTC da advanced

- advanced ACL — manbaga imkon qadar yaqin
- basic ACL — faqat manba IP; juda yaqin qo'yilsa kerakli trafik ham yopiladi
- Cheklov: L2–L4 sarlavha. Dinamik seans — ASPF, foydalanuvchi — 802.1X / AAA

Xulosa

- ACL = permit/deny qoidalar ro'yxati; birinchi mos qoida ishlaydi
- Basic 2000–2999 · Advanced 3000–3999 · Ethernet frame 4000–4999
- wildcard: 0 = tekshiriladi, 1 = e'tiborsiz
- packet-filter inbound / outbound — har yo'nalishga alohida
- Mos kelmagan paket — default action (default: permit)

Tekshirish

1. Advanced ACL raqami? — 3000–3999
2. 0.0.3.255 nechta bitni tekshiradi? — birinchi 22 bit
3. auto tartibda qaysi qoida oldin? — diapazoni torroq
4. Default filtering action? — permit